MILITARY SPECIFICATION

DISTRIBUTED ISOLATION MATERIAL TYPE R-50-21/64

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense

1 SCOPE

1 1 Scope This specification covers Navy type R-50-21/64 homogeneous rubber pad for use in machinery and equipment mounts to isolate vibration and attenuate shock

1 2 Classification The above-described rubber pad shall be composed of material generally termed distributed isolation material (DIM) and shall be referred to as DIM pad The numbers and letters in Navy type R-50-21/64 DIM pad shall indicate the following (see 6.2).

(a) The letter "R" denotes the general class of rubber in accordance with ASTM D 1418.
(b) The number "50" designates the design load rating in pounds per square inch (lb/in^2) of projected pad area The 50 lb/in^2 measurement is equivalent to 0.345 megapascal (MPa) or 345 kilonewtons per square meter (kN/m^2)
(c) The second number (21/64) denotes the design parameter, in this case, the nominal DIM pad thickness in inches (8.3 millimeters (mm))

2. APPLICABLE DOCUMENTS

2 1 Government documents

2 1 1 Specifications and standards The following specifications and standards form a part of this document to the extent specified herein Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

AMSC N/A FSC 5330
DISTRIBUTION STATEMENT A Approved for public release; distribution is unlimited

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:13Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-L-10547 - Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof Flexible

STANDARDS

FEDERAL

FED-STD-601 - Rubber· Sampling and Testing.
FED-STD-313 - Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-407 - Visual Inspection Guide for Rubber Molded Items

(Unless otherwise indicated, copies of federal and Military specifications and standards are available from the Naval Publications and Forms Center, (ATTN NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099)

2 2 Non-Government publications The following documents form a part of this document to the extent specified herein Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6 2)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 395 - Standard Test Methods for Rubber Property - Compression Set (DoD adopted)
D 412 - Standard Test Methods for Rubber Properties in Tension (DoD adopted)
D 471 - Standard Test Method for Rubber Property - Effects of Liquids.
D 573 - Standard Test Method for Rubber - Deterioration in an Air Oven (DoD adopted)
D 792 - Standard Test Methods for Specific Gravity (Relative Density) and Density of Plastics by Displacement (DoD adopted)
D 1149 - Standard Test Method for Rubber Deterioration - Surface Ozone Cracking in a Chamber. (DoD adopted)
D 1229 - Standard Test Method for Rubber Property - Compression Set at Low Temperatures (DoD adopted)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:13Z
Check the source to verify that this is the current version before use.

ASTM (Continued)

D 1418 - Standard Practice for Rubber and Rubber Latices - Nomenclature.

D 2240 - Standard Test Method for Rubber Property - Durometer Hardness (DoD adopted)

D 3182 - Standard Practice for Rubber - Materials, Equipment, and Procedures for Mixing Standard Compounds and Preparing Standard Vulcanized Sheets. (DoD adopted)

D 3951 - Standard Practice for Commercial Packaging. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4 3

3 2 Material

3 2.1 Composition The pads shall be fabricated of oil-resistant compounds utilizing polymerized chloroprene or nitrile rubber as the basic material in accordance with ASTM D 1418 The use of any rubber which gives off toxic, offensive, or damaging volatiles at operating temperatures up to 125 degrees Fahrenheit (°F) 52 degrees Celsius (°C) shall be prohibited No acceptable toxicity requirements exist for temperatures above 125°F.

3.2.2 Recovered materials Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3 3 Appearance. Rubber DIM pads shall be free of visual defects in accordance with MIL-STD-407 as specified in 4.5.2.

3 3.1 Surface pattern Unless otherwise specified (see 6 2), the surface pattern shall be at the option of the manufacturer

Source: https://assist.dla.mil - Downloaded: 2026-10-20T01:13Z
Check the source to verify that this is the current version before use.

3 4 Dimensions Dimensions of the pad shall be measured in accordance with FED-STD-601 as specified in 4.5.2.1. Dimensions of surface pattern shall conform to the drawing dimensions with a tolerance of plus or minus 0.010 inch (0 26 mm) (see 3.7 1) Dimensions and distributions of voids of the surface pattern shall not preclude cutting the DIM pad into annular (machinery-mounting) pads with walls as narrow as 1/2 inch (13 mm). The pads shall have two hardness-proof buttons molded onto opposite edges and lying outside the patterned portion of the pad and securely attached to the parent pad by flash or runners. The buttons shall be of nominal 1.25 inches in diameter by 21/64 inch thick (32 by 8 3 mm thick) Unless otherwise specified (see 6.2), the rubber DIM pad shall be 18 by 18 $\pm$ 1/4 inches (457 by 457 $\pm$ 6 mm) excluding the proof buttons Thickness of the rubber DIM pad shall be 0.328 $\pm$ 0.016 inch (8.3 $\pm$ 0.4 mm)

3 5 Properties of rubber stock. Samples of the rubber compound used in the pads, as specified in 4.3.1, shall conform to the requirements shown in table I when tested as specified in 4.5 3.

TABLE I Requirements for rubber stock

Physical properties	Requirements	Test procedure
Initial properties		
Tensile strength (min), lb/in^2 (MPa)	1800 (11 7)	4.5 3 1
Ultimate elongation, percent	350	4.5.3 1
Properties after oven aging		
Tensile strength, lb/in^2 (MPa)	1700 (11.7)	4.5.3.2
Ultimate elongation, percent	300	4 5 3.2
Properties after immersion in test fluid		
Tensile strength, lb/in^2 (MPa)	1700 (11.7)	4.5 3 3
Ultimate elongation, percent	300	4.5 3 3
Volume change, percent	plus or minus 10	4.5.3.4
Cold compression set (max), percent	35	4.5.3 5

3.6 Performance requirements.

3 6.1 Deflection at rated load The deflection of the DIM pads at the 50 lb/in^2 (0.345 MPa) load point shall not exceed 65 mils (1.65 mm) when tested as specified in 4.5.4.1. Compressed height at the 50 lb/in^2 (0.345 MPa) point shall be not less than 0 247 inch minimum (6.27 mm).

3 6.2 Resonant frequency before compressive strength test The resonant frequency of vibration of the rubber pad with supported load shall not exceed the values specified in table II when measured by the procedures specified in 4.5.4.2

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:13Z
Check the source to verify that this is the current version before use.

TABLE II Resonant frequency requirement.

Load, lb/in^2	Resonant frequency, Hz
25 (0.172 MPa) 50 (0.345 MPa)	27.0 22 0

3 6 3 Compressive strength. The pad shall not rupture when subjected to the compressive strength test specified in 4.5.4.3.

3.6.4 Resonant frequency after compressive strength test. The resonant frequency of the rubber pad after being subjected to the compressive strength test (see 3 6.3) shall be changed by not more than plus or minus 1 hertz (Hz) when retested in accordance with 4 5.4.4 at a supported load of 50 lb/in^2 (0.345 MPa).

3.6 4 1 Methods for determining resonant frequency. Two different measurement methods for determining resonant frequency of pad samples under specified loads may be used. Either method is acceptable and will provide comparable results when equipment and procedures are as specified in 4.5 4 4 1

3.6.5 Drift. The drift of the rubber pad shall not exceed 8 mils (0.20 mm) when tested in accordance with 4.5.4.5

3.6 6 Hardness The hardness of the pads shall be determined by the procedure as specified in 4.5 5. The hardness value found during first article inspection (see 3.1) of the contractor's DIM pads shall be recorded as a requirement with a tolerance of plus or minus 5 Shore A hardness units for subsequent quality conformance testing of this contractor's pads.

3 6.7 Specific gravity. The specific gravity of the rubber in the DIM pad shall be determined by the procedure as specified in 4 5.6 The specific gravity found during first article inspection (see 3 1) of the contractor's DIM pads shall be recorded as a requirement with a tolerance of plus or minus 0.02 unit for subsequent quality conformance testing of this contractor's pads.

3 6.8 Ozone resistance. The rubber pad shall not be cracked by exposure to ozonized air when tested in accordance with 4.5.7.

3.7 Requirements for finished pads.

3 7.1 Resonant frequency Resonant frequency of the pad shall not exceed the values of table II when tested in accordance with 4 5 4 2.

3.7.2 Deflection at rated load. The deflection of the pads when tested at their rated loads shall be within the limits specified in 3.6.1.

3.7.3 Drift. The drift test shall be in accordance with 4.5.4.5

3.7.4 Strength. The pad shall be subjected to the compressive test specified in 4 5.4.3 and shall not show evidence of rupture.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:13Z
Check the source to verify that this is the current version before use.

3.8 Drawings. (See 6.3).

3.9 Identification marking. DIM pads shall be identified by markings on the shipping containers in accordance with MIL-STD-129 (see 5.3)

3.10 Toxicity. The pads shall have no adverse effect on the health of personnel when properly used for its intended purpose. The pads shall contain no components which produce noxious vapors in such concentrations as to be an annoyance to personnel during formulation or use under conditions of adequate ventilation while exercising caution to avoid prolonged contact with the skin and while observing Occupational Safety and Health Administration (OSHA) guidelines Questions pertaining to the toxic effects shall be referred by the contracting activity to the Naval Medical Command (NAVMEDCOM), Washington, DC, who will act as a medical advisor to the contracting activity.

3.10.1 Material safety data sheet (MSDS). The contracting activity shall be provided a material safety data sheet at the time of contract award. The MSDS shall be provided in accordance with the requirements of FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification (see 6 5).

4 QUALITY ASSURANCE PROVISIONS

4 1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4 1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4 3 First article inspection. First article inspection shall consist of the examination specified in 4 5.2 and tests as specified in 4.5.3 and 4.5 4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:13Z
Check the source to verify that this is the current version before use.

4 3.1 Sampling for first article. The following samples shall be drawn for first article inspection and testing:

(a) Three sheets of the rubber stock, 6 by 6 inches by 0.080 ± 0.010 inch (150 by 150 by 2 mm) shall be prepared in accordance with ASTM D 3182. The sheets shall be certified to have been made from the same rubber compound as the rubber pads with an equivalent cure. Grain direction shall be marked to facilitate tension testing.
(b) Three rubber DIM pads with attached hardness-proof buttons representative of production under the contractor's identifying number and as specified in 3.2
(c) Four compression set buttons having 1.14 ± 0.02 inch (29.0 ± 0 5 mm) diameters with thicknesses of 0.50 ± 0.02 inch (13.0 ± 0.2 mm) prepared in accordance with ASTM D 395. The buttons shall be certified to have been made from the same rubber compound as the rubber pads and to have an equivalent cure.

4 3 2 First article examination. Should any of the pads submitted for first article examination or measurement exhibit major or minor defects as specified in 4.5.2 and 4.5.3 or if the pads fail the tests as specified in 4 5 4 this shall be cause for rejection.

4 4 Quality conformance inspection Quality conformance inspection shall consist of the examination, measurements, and tests as specified in 4.5 2 and table III and shall be performed in the presence of a qualified NAVSEA representative or a representative of the Command or agency concerned

TABLE III. Quality conformance tests

Property	Test procedure
Properties of rubber stock	
Tensile strength	4 5.3.1
Ultimate elongation	4.5.3.1
Specific gravity	4.5.6
Properties of pads	
Deflection under load	4.5.4.1
Resonant frequency	4 5 4 2
Hardness	4.5.5

4.4.1 Lot. A lot for quality conformance inspection shall consist of all rubber DIM pads produced in one plant under essentially the same conditions and offered for delivery at one time In each case the number of DIM pads shall be the lot size.

4 5 Procedures for examination, measurements, and tests.

4.5 1 Temperature of tests Unless otherwise specified, every sample shall be conditioned for at least 4 hours at 80 ± 9°F (26 7 ± 5°C) before being tested, and the tests shall be performed at this temperature

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:13Z
Check the source to verify that this is the current version before use.

4 5 2 Visual and dimensional examination. Each of the rubber DIM pads submitted for first article inspection (see 4.3.1) or sampled for quality conformance inspection shall be examined for major and minor defects in accordance with MIL-STD-407 to determine conformance to 3.3. Surfaces of the DIM pad shall be classified as location I in accordance with MIL-STD-407 for the purposes of this examination. Bloom of the surface of the pad shall not be considered a defect.

4.5.2.1 The length, width, and thickness of the DIM pad and the dimensions of the surface design shall be measured in accordance with methods 2001, 2011, 2021, 2031, 2121, 2321 and 2361 of FED-STD-601, as applicable. This shall be done to define the surface design on the DIM pad during first article inspection and to determine conformance to 3.4 for quality conformance. During quality conformance measurements, any deviation of the surface design defined for first article (see 3.1), after allowing for the 0.010 inch (0.26 mm) tolerance (see 3 4), shall be regarded as a major defect as shown in table III to determine acceptance or rejection Any deviation of the length, width, and thickness of the DIM pad from the specified dimensions, after allowing for tolerances (see 3 4 and 6.2) shall be regarded as a major defect.

4.5.3 Tests on special rubber samples. The properties of the 0.080 inch (2 mm) thick samples and compression set buttons (see 4.3.1) representing the rubber pads shall be evaluated to determine conformance to the requirements of 3.5.

4 5.3.1 Initial tensile strength and ultimate elongation The initial tensile strength and ultimate elongation of the rubber stock shall be determined in accordance with ASTM D 412 using Die C samples.

4.5 3 2 Tensile strength and ultimate elongation after oven aging. The tensile strength and ultimate elongation of the rubber stock shall be determined after oven aging in accordance with ASTM D 412 and D 573 except that the aging period shall be 70 ± 1/4 hour at a temperature of 212 ± 2°F (100 ± 1°C) Die C samples shall be used.

4.5 3 3 Tensile strength and ultimate elongation after oil immersion The tensile strength and ultimate elongation of the rubber shall be determined after immersion in ASTM No. 1 oil for 46 ± 1/4 hours at a temperature of 158 ± 2°F (70 ± 1°C) in accordance with ASTM D 412 and D 471 using Die C samples The tensile strength reported shall be based on the original cross section of the samples

4 5.3.4 Volume change after oil immersion. The change in volume of the rubber shall be determined after immersion in ASTM No. 1 oil for 46 ± 1/4 hour at a temperature of 158 ± 2°F (70 ± 1°C) in accordance with ASTM D 471

4.5.3.5 Cold compression set. Cold compression set shall be determined in accordance with ASTM D 1229, except that the exposure shall be 30 ± 2°F (minus 1 ± 1°C) for 94 ± 1/2 hours Compression set shall be determined 30 minutes after release from the jig

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:13Z
Check the source to verify that this is the current version before use.

4.5.4 Destructive tests on pads.

4.5.4.1 Deflection at rated load. Three samples, each 2-3/4 ± 1/16 inch (70 mm) in diameter with a centrally located hole 1-1/8 ± 1/16 inch (28.6 mm) in diameter shall be prepared from the DIM pad. The samples shall be cut from the pads by means of a rotating cutting tool similar to that shown on figure 1 They shall be lubricated with soapy water so that a smooth-cut surface having square edges is obtained The exact projected surface area of each sample shall be calculated from measurement of these dimensions to the nearest 1/64 inch (0.4 mm).

4.5.4.1 1 The deflection under load of each sample in turn shall be determined using a universal testing machine capable of measuring load in units not larger than 5 pounds (22 newtons) and deflection in units not larger than 0.005 inch (0.13 mm). Each sample shall be subjected to four loading and unloading cycles up to a peak load of 75 lb/in^2 (0.517 MPa). For the first three cycles the deflection rate shall not exceed 0.2 inch per minute (in/min) (5 millimeters per minute (mm/min). For the fourth cycle, the rate of loading and unloading shall not exceed 0 05 in/min (1.3 mm/min). The deflection of the sample in inches measured during the loading phase of the fourth cycle shall be plotted versus load, and the deflection produced by a load of 50 lb/in^2 (0.345 MPa) shall be read from the curve. The average of the values obtained on the three samples at this load shall be used to determine conformance to 3.6.1.

4.5 4.1 2 The loaded height of each of the three samples at 50 lb/in^2 (0 345 MPa) load shall be measured. This measurement shall be made at the same time as the deflection-load test (see 4.5.4.1 1), using a dial gauge graduated in thousandths of an inch, and shall be taken during the loading phase of the fourth cycle The three values obtained shall be averaged to determine conformance with loaded height criteria of 3 6 1

4.5.4.2 Resonant frequency before compressive strength test. The three samples used for the deflection-under-load test (see 4.5.4.1) shall be used for the resonant frequency test The resonant frequencies of vibration of each sample shall be determined by one of the methods of 4.5 4.4 1 while supporting purely mass loads of 25 lb/in^2 (0.172 MPa) and 50 lb/in^2 (0.345 MPa) respectively. The clean, dry test sample shall be compressed between clean, dry, polished steel platens. The vibration excursion (double amplitude) shall not exceed 0.01 inch (0.25 mm) during the test. The averages of the resonant frequency values obtained for the three samples at each load shall be used to determine conformance to 3.6.2.

4.5.4.2 1 To ensure acceptable performance of larger DIM pads having patterns not previously tested, an additional sample 3-3/4 ± 1/16 inch (95 mm) in diameter with a centrally located 1-1/8 ± 1/16 inch (29 mm) hole shall be tested for resonant frequency at 25 lb/in^2 (0 172 MPa). Results of this check shall also be within the limit specified in 3.6.2.

4.5.4.3 Compressive strength Each of the three 2-3/4 inch (70 mm) samples tested in 4.5.4.2 shall be subjected to a load of 10,000 lb/in^2 (69 MPa) in a universal testing machine. The rate of loading and unloading shall not exceed 0 05 in/min (1 3 mm/min) with no dwell time at maximum load. Maximum deflection shall be observed and the samples examined to determine conformance to 3.6.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:13Z
Check the source to verify that this is the current version before use.

4 5 4 4 Resonant frequency after compressive strength test. The resonant frequency of vibration of each of the three samples subjected to the compressive strength test of 4 5.4.3 shall be determined while they support a load of 50 lb/in^2 (0 345 MPa). The test conditions specified in 4.5 4 2 shall be observed The average of the resonant frequency values obtained for three specimens shall be used to determine conformance to 3.6.4.

4.5.4.4 1 Determination of resonant frequency.

4 5.4 4 1 1 Resonant methods These methods shall permit the direct measurement of resonant frequency of loaded pad specimens for test machines meeting certain stiffness criteria A suitable arrangement for performing this test and the associated electronic circuitry is shown on figures 2 and 3. The pad is supported by a test stand of high stiffness (low compliance) and compressed to the desired unit loading with weights suspended from a steel rod The system is excited by an electromagnetic shaker attached to the bottom of the rod. The impedance head senses both the dynamic force applied to the system as well as its motion. The instrumentation shown on figure 3 amplifies and ratios the two signals to provide a single voltage proportional to the ratio of input velocity to force, that is, the drive point mobility of the system. The procedure shall require the application of a sinusoidal vibration to the system, slowly sweeping upward in frequency while not exceeding a vibration amplitude (peak-to-peak) of 0 01 inch (0.25 mm). At the resonant frequency of the system the mobility ratio is a maximum

4.5 4.4.1.1 1 In lieu of using an impedance head and ratioing instrumentation, separate force and motion transducers and readout instruments may be used The procedure shall again be to sweep upward in frequency with sinusoidal vibration, this time maintaining constant input force amplitude and determining the frequency at which the signal from the motion transducer is a maximum Whichever method is used, it shall be essential that the input force to the system be accounted for in either of the previous ways so that the measured resonant frequency will be independent of the properties of the vibration exciter

4.5 4 4 1 2 Nonresonant methods. These methods shall be based on the transmitted force principle. One such system is the servo-controlled electro-hydraulic test machine which subjects a rubber sample to given loads and superimposes upon those loads a sinusoidal force. A schematic diagram of an electro-hydraulic test machine is shown on figure 4. Electronic circuitry controlling the servo valve permits the operator to independently vary the excitation frequency, vibration amplitude, and load applied to the sample. The dynamic spring rate is determined by measuring the dynamic force transmitted through the sample, the vibration amplitude across the sample, and the phase angle between these quantities in accordance with the following equation·

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:13Z
Check the source to verify that this is the current version before use.

$$Cx + Kx = F$$

Where.

C = Viscous damping coefficient, pound second per inch (newton second per meter)

F = F(t) = Sinusoidal transmitted force, pound force (newton)

K = Dynamic spring rate of sample pounds per inch (newton/meter)

x = x(t) = displacement across sample, inch (meter)

$\dot{x} = \frac{dx}{dt}$ = velocity across sample, inches per second (meters per second)

Although the more exact equation also includes a mass term, it may be omitted with negligible error for small specimens and low excitation frequencies. To determine conformance to 3 6.2 and 3.6.4, the dynamic spring rate thus obtained may be used to calculate a resonant frequency by considering the force load on the sample to be entirely due to an inertial (mass) load as follows

English system	Metric system
$f = 3\ 127\sqrt{\frac{K}{w}}$	$f = 0.159\sqrt{\frac{K}{m}}$
where K = dynamic spring rate, pounds per inch	where K = dynamic spring rate, newtons per meter
f = resonant frequency, Hz	f = resonant frequency, Hz
w = supported weight, pound	m = supported mass, Kg

4 5 4 4.1.3 Calibration of test systems Any system used in the resonant and nonresonant testing of pads to determine the resonant frequency shall be calibrated before use Steel coil springs of known spring rate may be used for calibration purposes to ensure that the equipment behaves correctly with changes in stiffness, and in a linear manner with variations of excitation amplitude and frequency. Care shall be taken to insure that the compliance of the test equipment does not result in erroneously low values for resonant frequency and that the error due to this effect does not exceed 1 percent of the correct resonant frequency value The test equipment should be at least a factor of 100 times stiffer than the sample being tested

4 5 4 5 Drift One sample, 2-3/4 ± 1/16 inch (70 mm) in diameter with a centrally located hole 1-1/8 ± 1/16 inch (29 mm) in diameter, shall be prepared from the DIM pad The exact surface of the sample shall be calculated from measurement of these dimensions to the nearest 1/64 inch (0.4 mm). The clean, dry sample shall be loaded in compression to 50 lb/in^2 (0.345 MPa) between clean, dry, polished steel platens using a deadweight. The height of the sample in mils

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:13Z
Check the source to verify that this is the current version before use.

shall be measured 1 hour after application of the load and 336 hours (2 weeks) after application of the load. The difference in the two readings in mils shall be reported as drift and used to determine conformance to 3 6.5 An apparatus similar to the one shown on figure 2 shall be used for this test after removal of the vibration exciter.

4.5.5 Hardness test. Hardness of the DIM pads shall be determined on hardness proof buttons of nominal 1 25 inches in diameter by 21/64 inch thick (32 by 8.3 mm thick) molded on opposite edges of the pad as specified in 3.4 Measurements shall be made in accordance with ASTM D 2240 using a Shore A durometer and 3-second readings. A single hardness value consisting of the average of the readings taken shall be obtained for each button. These values, two for each pad tested, shall be used to determine conformance to 3 6 6 Failure of either hardness button to conform to 3.6.6 shall be regarded as a major defect in determining acceptance or rejection for quality conformance or during first article testing (see 4 3 2)

4 5.6 Specific gravity. The specific gravity of the rubber compounds used for the DIM pads shall be determined on samples cut from 6 by 6 by 0.080 inch sheets using ASTM D 792. The average of three determinations shall be used to determine conformance with 3 6 7

4.5.7 Ozone resistance Three samples, each 2-3/4 $\pm$ 1/16 inch (70 mm) in diameter with a centrally located 1-1/8 $\pm$ 1/16 inch (29 mm) diameter hole shall be prepared from the pad The clean, dry samples shall be compressed between clean, dry steel plates and held at a deflection corresponding to a load of 50 lb/in^2 or 0 345 MPa (see 4.5 4 1) The compressed samples shall be conditioned for 16 to 18 hours at a temperature of 100 $\pm$ 2°F (38 $\pm$ 1°C) in an oven in accordance with ASTM D 573 The compressed samples shall then be exposed to air at a temperature of 100 $\pm$ 2°F (38 $\pm$ 1°C) containing ozone at a concentration of 1.00 $\pm$ 0 05 parts per million by volume The time of exposure shall be 166 $\pm$ 1/2 hour The exposure shall be carried out in a chamber conforming to ASTM D 1149, and the measurement of the ozone concentration shall be in accordance with ASTM D 1149 After exposure, the edges of the compressed samples shall be examined for cracks in the rubber to determine conformance to 3.6.8. A magnifying lens of at least 7X power shall be used for the examination

4 6 Toxicity To determine conformance to the requirements of 3 10, the manufacturer of the material shall disclose the formulation of his product to the Naval Medical Command, MEDCOM-242, Washington, DC 20372. The disclosure of proprietary information, which shall be held in confidence by the Naval Medical Command, shall include. the name, formula, and approximate percentage by weight and volume of each ingredient in the product; the results of any toxicological testing of the product; identification of its pyrolysis products; and any such other information as may be needed to permit an accurate appraisal of any toxicity problem associated with the handling, storage, application, use, disposal, or combustion of the material

4 7 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:13Z
Check the source to verify that this is the current version before use.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5 1 Preservation Preservation shall be level A, C or commercial, as specified (see 6 2)

5 1 1 Level A DIM pads of the same class, load rating and design shall be preserved method IC in accordance with MIL-P-116 as follows:

(a) Sufficient pads weighing between 60 and 70 pounds (27 to 32 kilograms (kg)) shall be packed flat in a snug fitting fiberboard container conforming to PPP-B-636 grade V, type optional. Containers shall be furnished with a reinforcing liner and top and bottom pads.
(b) To prevent sticking, an interleaving material shall be placed between each pad prior to packing in the container.
(c) Containers shall be closed in accordance with method V of the appendix to the container specification.

5 1.2 Level C DIM pads of the same class, load rating and design shall be preserved and packed as specified herein for level A, except that containers shall be of the domestic type and the grade shall be minimum 350 To minimize ozone attack, the container shall be closed method V of the appendix to the container specification

5 1 3 Commercial DIM pads shall be packaged in accordance with ASTM D 3951

5 2 Packing. Packing shall be level A, B, C or commercial, as specified (see 6.2)

5 2 1 Level A DIM pads preserved as specified in 5.1 shall be packed in wood-cleated plywood or nailed wood boxes conforming to PPP-B-601 overseas type and PPP-B-621 class 2 respectively with selection of the container at the contractor's option Caseliners for boxes containing DIM pads preserved level A will not be required Boxes containing DIM pads preserved level C or commercial will require caseliners conforming to MIL-L-10547. Boxes shall be closed and strapped in accordance with the applicable box specification or appendix thereto The gross weight of boxes shall not exceed 200 pounds (90 72 kg).

5.2.2 Level B. DIM pads preserved as specified for level A in 5.1 will require no further packing. DIM pads preserved as specified for level C or commercial in 5.1, shall be overpacked in wood containers specified in 5.2 1 except that the containers may conform to the domestic type or class, or in fiberboard containers conforming to PPP-B-636 class weather-resistant or PPP-B-640 weather-resistant. The contractor shall have the option as to container selection Box closure and reinforcement shall conform to the applicable box specification or appendix thereto, except that method V closure shall be applicable for PPP-B-636 boxes The gross weight of wood boxes shall not exceed 200 pounds (90 72 kg), of fiberboard for the gross weight specified for the applicable box used

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:13Z
Check the source to verify that this is the current version before use.

5 2 3 Level C DIM pads preserved as specified for level A and level C in 5 1 will require no further packing DIM pads preserved commercial as specified in 5 1 shall be overpacked as specified for level B herein except that containers used may be of the domestic type or class and method I closure shall apply for PPP-B-636 boxes

5 2 4 Commercial. DIM pads preserved as specified in 5.1 shall be packed in accordance with ASTM D 3951.

5 3 Marking In addition to the special marking required herein, or by the contract or order (see 6 2), interior packs and exterior shipping containers for levels A, B and C shall be marked in accordance with MIL-STD-129 Commercial packs and shipping containers shall be marked in accordance with ASTM D 3951

5 3.1 Special marking. In addition to the marking required in 5.3, shipment marking information shall include Navy type number and date of cure

6 NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory)

6 1 Intended use The oil and ozone-resistant DIM pad covered by this specification is intended for use in machinery and equipment mounts in order to isolate vibration and to attenuate shock, where temperatures do not exceed 125°F (52°C)

6 2 Acquisition requirements Acquisition documents must specify the following

(a) Title, number, and date of this specification
(b) Type and class required (see 1 2)
(c) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2 1 1 and 2 2)
(d) When first article inspection is required (see 3 1)
(e) When surface pattern is specified (see 3.3 1).
(f) Dimensions, when other than specified (see 3.4 and 4.5.2 1)
(g) Levels of preservation and packing required (see 5.1 and 5.2)
(h) When special marking is required (see 5 3).

6 3 Consideration of data requirements The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/ provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27 475-1 exempts the requirement for a DD Form 1423

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:13Z
Check the source to verify that this is the current version before use.

Reference paragraph	DID number	DID title	Suggested tailoring
3 8	DI-E-7031	Drawings, engineering associated lists	Level 3
4.3	DI-T-4902	First article inspection report	--
4 4	DI-T-5329	Inspection and test reports	--

The above DIDs were those cleared as of the date of this specification The current issue of DoD 5010 12-L, Acquisition Management Systems and Data Requirements Control list (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6 4 First article When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first _____ production items, a standard production item from the contractor's current inventory (see 3 1), and the number of items to be tested as specified in 4.3 The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6 5 Material safety data sheet (MSDS). Contracting officers must identify those activities requiring copies of MSDSs Additional required Government information is contained in FED-STD-313. In order to obtain the MSDS, FAR clause 52.223-3 must be in the contract

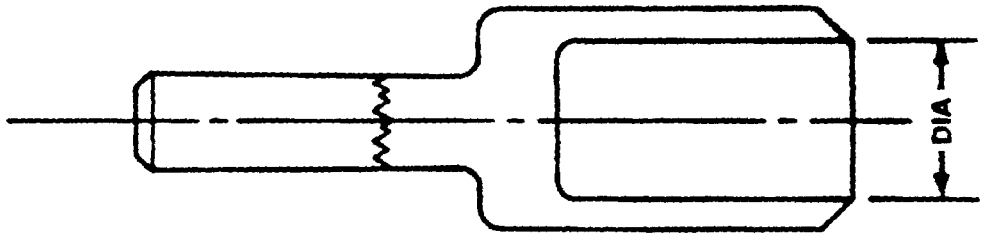
6 6 First article inspection report The manufacturer should include a detailed description of his test equipment and procedure used to determine resonant frequency (see 4.5.4.4.1.3) in the first article inspection report

6 7 Subject term (key word) listing

Compressive strength
Hardness
Pad samples
Resonant methods
Specific gravity

Preparing activity:
Navy - SH
(Project 5330-N096)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:13Z
Check the source to verify that this is the current version before use.

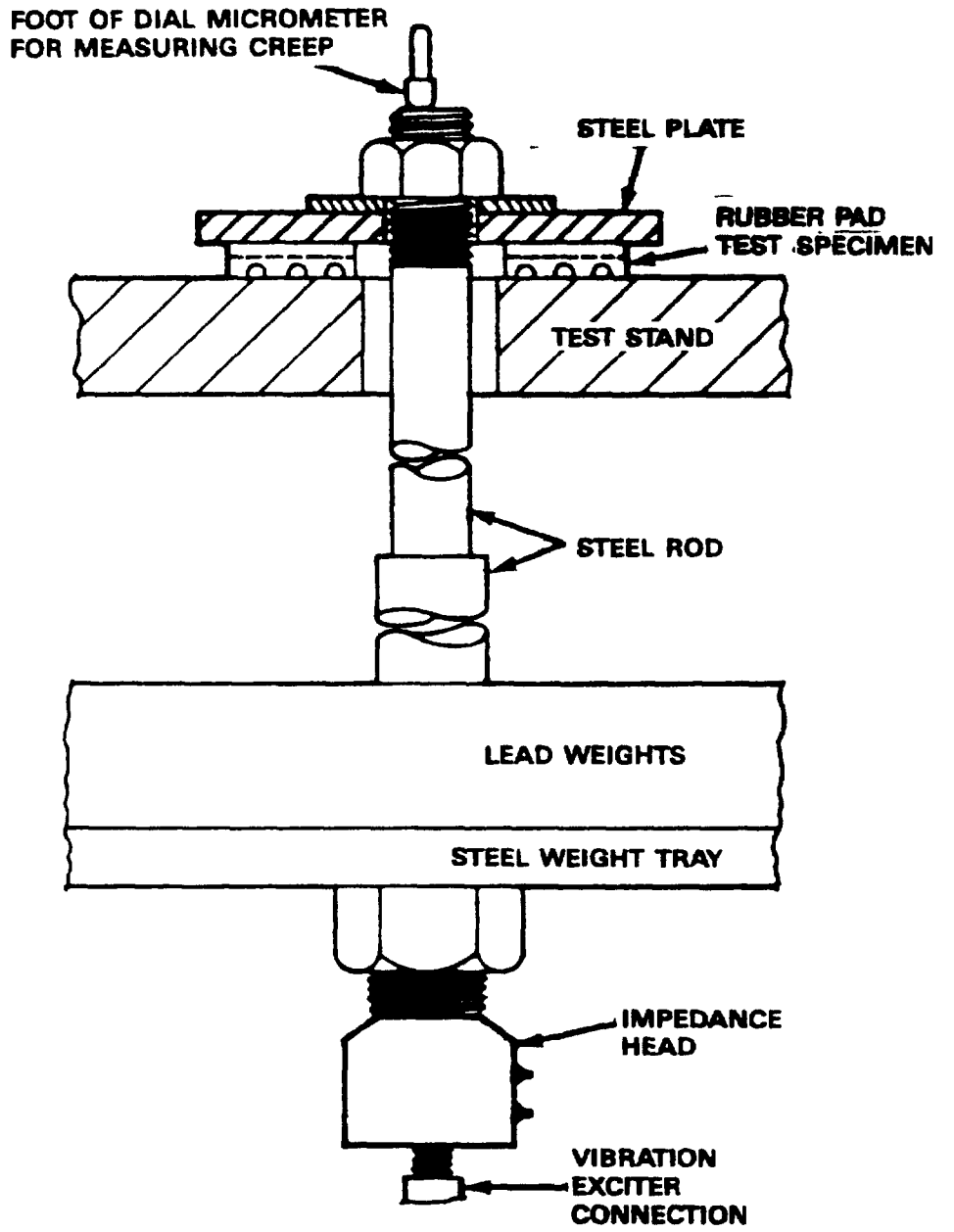


CUTTING TOOL	DIAMETER
FOR SPECIMEN OUTSIDE DIAMETER	2.755±0.005 IN. (69.98±0.13mm)
FOR SPECIMEN INSIDE DIAMETER	1.130±0.005 IN. (28.70±0.13mm)

SH 13202315

FIGURE 1 Cutting tools for pad specimens.

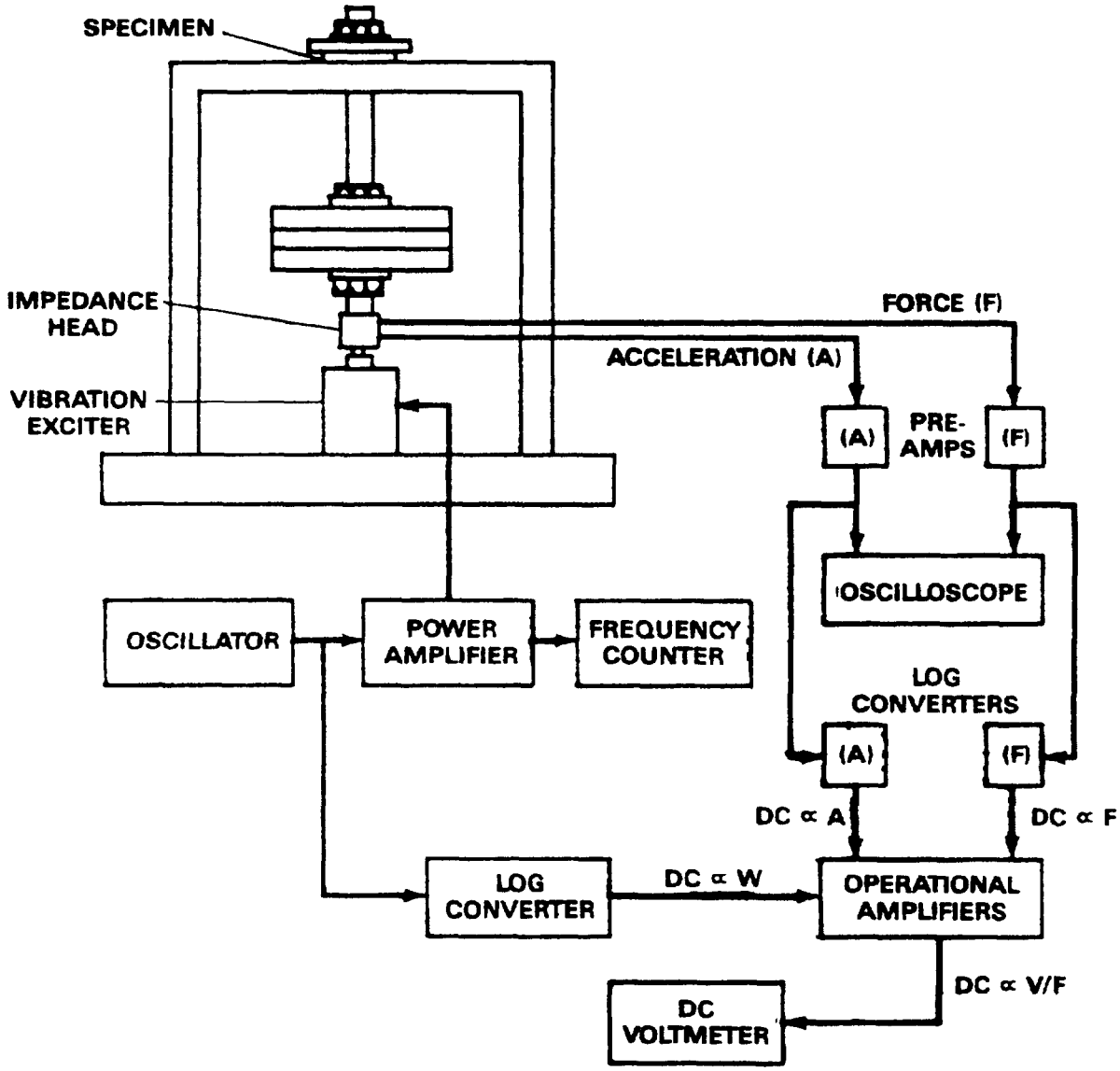
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:13Z
Check the source to verify that this is the current version before use.



SH 13202316

FIGURE 2. Typical arrangement for measurement of resonant frequency by the suspended mass method and for measuring creep (after disconnecting vibration exciter connection).

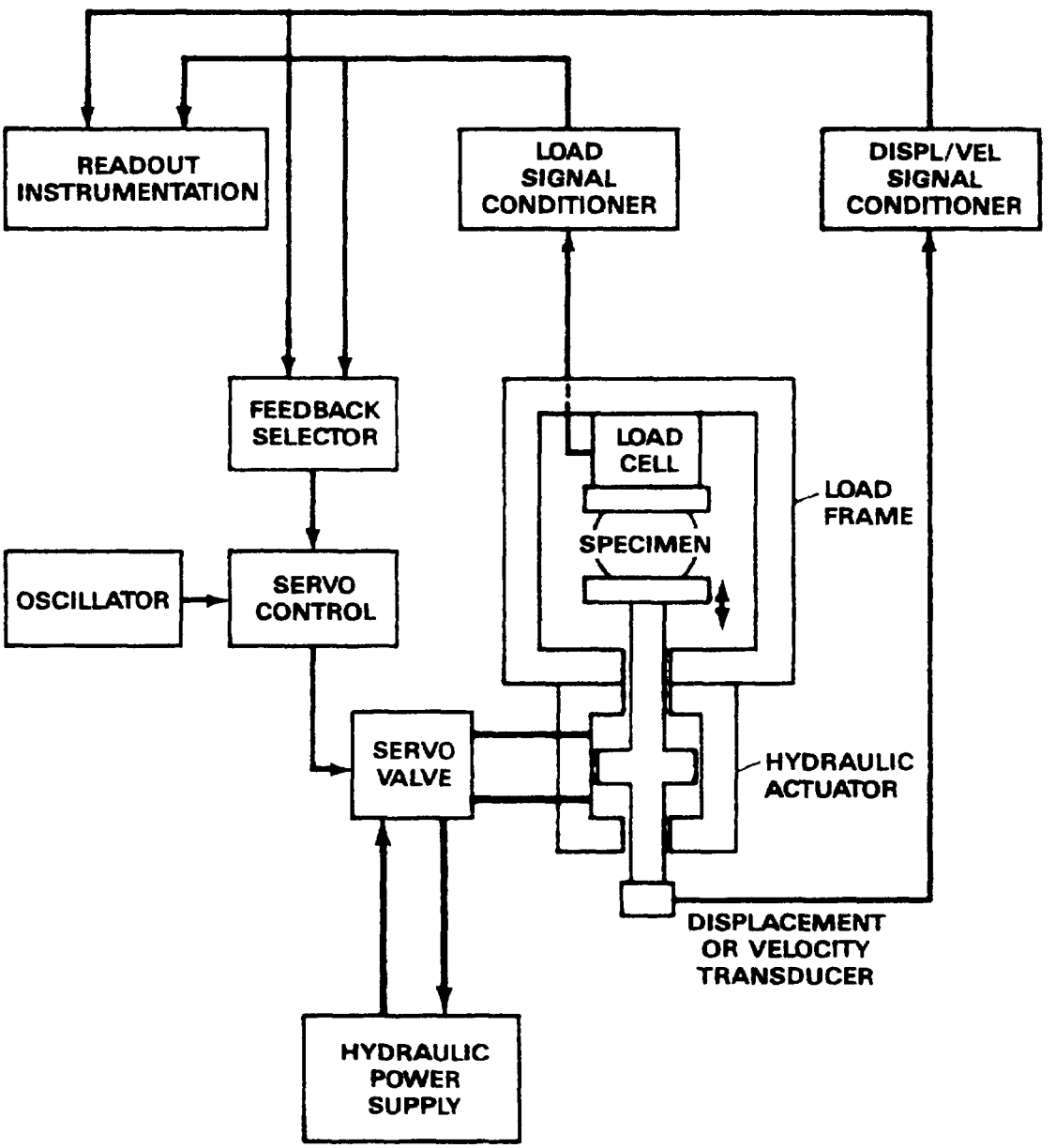
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:13Z
Check the source to verify that this is the current version before use.



SH 13202317

FIGURE 3 Typical electronic instrumentation for measurement of resonant frequency by the suspended mass method.

Source: https://assist.dla.mil - Downloaded: 2026-10-20T01:13Z
Check the source to verify that this is the current version before use.



SH 13202318

FIGURE 4 Schematic diagram of major components of electrohydraulic closed-loop test machine.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:13Z
Check the source to verify that this is the current version before use.

(TO DETACH THIS FORM CUT ALONG THIS LINE)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-D-24709(SH)	DISTRIBUTED ISOLATION MATERIAL TYPE R-50-21/64

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	8. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	9. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

Source: https://assist.dla.mil – Downloaded: 2025-10-20T01:13Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS. In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE. This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

OMMANDER
AVAL SEA SYSTEMS COMMAND (SEA 55Z3)
EPARTMENT OF THE NAVY
ASHINGTON, DC 20362-5101

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12503 WASHINGTON D C
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDER
NAVAL SEA SYSTEMS COMMAND (SEA 55Z3)
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20362-5101

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:13Z
Check the source to verify that this is the current version before use.